XINGQIANG CHEN

Hangzhou, China
+86 150-0133-4125 ◇ joy6677@outlook.com ◇ github.com/chenxingqiang

OBJECTIVE

Looking for **FULL TIME** roles in **AI Accelerator Compiler / System Software** · **LLM Inference Systems** · **HW/SW Co-design**.

INTERDISCIPLINARY BACKGROUND

Physics and biophysics training across computational chemistry, molecular dynamics, and computer science. Contests: **CUMCM National First Prize** (2012); **MCM Meritorious Winner** (2013); **Second Prize**, CCF Open Source GPU Innovation Ecosystem Regular Season S1, GPU Operator Optimization Challenge (2025).

EDUCATION

Doctoral Studies in Biophysics and Condensed Matter Sep. 2014 - Jul. 2018
Xiamen University, Department of Physics and Engineering; protein dynamics and QM/MM simulation
Certificate of completion; degree defense in progress

B.S. in Theoretical Physics Sep. 2010 - Jul. 2014
Xiamen University, Department of Physics and Engineering

SKILLS AND PROJECTS

Languages: C++, Python (primary); CUDA, Java, Scala, Go.

Compiler / Toolchain: MLIR, LLVM backends and ISA bring-up, TVM, graph partitioning, operator fusion, cost modeling, codegen, auto-tuning.

Chip Software / Verification: Palladium emulation, GDB co-debug against simulation models, firmware and weight-map generation, operator-level metrics (Cycles/MACs).

Inference Systems: Transformers, KV-Cache, quantization, batching and scheduling, Triton kernels, FlashAttention, CUPTI/Perfetto profiling.

Platforms: PyTorch, TensorFlow, ONNX, Kubernetes; Spark, Flink, Hadoop.

Open Source & Releases (`github.com/chenxingqiang` unless noted):

- **TraceSmith** — GPU profiling & replay across CUDA/ROCm/Metal/MACA/Ascend; C++ core, Python bindings, on PyPI. `TraceSmith`
- **YiRage Persistent Kernel** — LLM megakernel compiler and runtime, multi-backend. `YiRage`
- **JAX-sklearn** — sklearn-compatible API on JAX; 13,058 tests passing; on PyPI. `jax-sklearn`
- **Secret-Learn** — Privacy-preserving ML: federated learning, secret sharing, split learning. `Secret-Learn`
- **YouTube-DNN-RecSys** — TensorFlow + Spark retrieval and ranking (60 stars). `YouTube-DNN-RecSys`
- **Research CLI** — Command-line AI tooling for research workflows. `iechor-research/research-cli`
- **Questions-Gen** — Math-problem generation (Qwen3-14B, GRPO, distillation); CLI and HF weights. `GenQuestion`

EXPERIENCE

Compiler & Toolchain Expert, Witmem Technology
Hangzhou, China
Compiler / Palladium Verification *Dec. 10, 2025 - Present*

- Owned the product-line compiler and Palladium emulation for a CIM chip: model compile, codegen, on-chip execution, and defect isolation on the emulator.
- Brought up software on Palladium: ran compiler output on the emulator and jointly debugged with GDB across the simulation model, silicon, and system software.

- Implemented **CIM-MLIR**: PyTorch/TensorFlow/ONNX lowered via TOSA/StableHLO to CPU, GPU, and CIM, covering frontend, graph partition, hardware backend, and runtime.
- Implemented cost-driven graph partition and CIMHigh/Mid/Low IR; used OR-Tools for Die/L2/L0 allocation; emitted burnable **config.bin**, weight **map.bin**, GPU PTX, and LLVM host code.
- Implemented a cross-device runtime: per-device executors, unified memory, compute/transfer overlap and triple buffering; C++/Python APIs with latency, utilization, and transfer-cost counters.
- Implemented **GRC-PIM**: drift-gated, GEMV-reusing residual calibration and soft-decision error correction for PIM; submitted to *npj Unconventional Computing* (see Publications).

Senior AI Compiler Engineer, Yizhu Tech Ltd.
Hangzhou, China
R&D Department *Sep. 2022 - Dec. 2025*

- Led **YiRage Persistent Kernel (YPK)**, compiling end-to-end LLM decode into a single megakernel; **1.2×–6.7×** end-to-end latency gain over a PyTorch eager baseline on Llama2-7B, single A100 (varies with batch and sequence length).
- Implemented fused operators (RMSNorm+Linear, GQA) and a persistent execution model, cutting kernel-launch and intermediate-tensor round trips; largest gains on the batch=1 decode path.
- Built the scheduling layer and integrated Triton and FlashInfer kernels; multi-GPU scaling validated up to **8** devices with NVSHMEM-backed tensors.
- Developed graph compilation and auto-tuning: kernel parameter search and KV-cache-friendly memory layout to reduce memory traffic.
- Filed **2** invention patents; delivered the stack into **2** datacenter customer PoCs.
- Owned the LLVM backend Target and complex-ISA support for the in-house SoC, driving backend standardization and the software ecosystem.
- Built model-conversion and operator-level evaluation: integrated TVM, ONNX, MLCommons, TensorFlow/PyTorch into **YZlite**, reporting Cycles/MACs and multi-chip benchmarks. `ReRAM-Labs/yzlite`

Privacy-preserving Algorithms Engineer, Trustbe
Hangzhou, China
IT Group *Jan. 2022 - Sep. 2022*

- Built a homomorphic-encryption federated learning platform for cross-institution modeling in finance, improving model quality over local-only baselines while keeping privacy constraints.
- Designed MPC protocols for joint training and inference of bank risk models, balancing protocol security against communication cost.
- Open-sourced **Secret-Learn**: sklearn-compatible API across federated learning, secret sharing, and split learning, JAX-accelerated and integrated with SecretFlow; on PyPI. `Secret-Learn`
- Filed **3** invention patents, all published: `CN114925786A`, `CN114997345A`, `CN115130681A`.

Staff Algorithms Engineer, BioMap
Beijing, China
Algorithm Engineering Group *Sep. 2021 - Dec. 2021*

- Built bioinformatics tooling and a unified data platform for large-scale sequence analysis, structure prediction, and functional annotation.
- Set up evaluation pipelines for protein structure prediction models (AlphaFold family) feeding downstream drug design.
- Two registered software copyrights (2021SR2004464, 2021SR1862397); open-sourced AlphaFold2 code walkthroughs (26 stars). `alphafold2-codecs`

Senior Algorithms Engineer, Yalla Group (NYSE: YALA)
Hangzhou, China
Data Center Engineering Group *Sep. 2020 - Sep. 2021*

- Rebuilt the ChatRooms recommender with multi-objective ranking; A/B tests showed significant gains in daily active time and 7-day retention (**42% → 61%**).
- Built Ludo risk control from real-time behavior features and ML models, materially reducing fraudulent refunds.
- Developed a card-game AI with deep RL for human-vs-bot play and matchmaking fill, improving off-peak match quality and retention.

Senior Algorithms Engineer, Gejia Network Tech — Hangzhou, China
Search & Recommendation Engineering Group — *Mar. 2019 - Sep. 2020*

- Designed multi-scenario ranking with collaborative filtering, deep ranking models, and a product knowledge graph; online experiments delivered double-digit relative CTR and CVR gains.
- Optimized the serving path with feature caching and candidate pruning, holding P99 latency under **50ms** through peak promotional traffic.
- Open-sourced YouTube-DNN-RecSys, a TensorFlow + Spark retrieval and ranking reference (60 stars). `chenxingqiang/YouTube-DNN-RecSys`

Machine Learning Engineer, Bianlifeng Tech — Beijing, China
Data Engineering Group — *Mar. 2017 - Mar. 2019*

- Built store-level sales forecasting across **50+** stores using gridded weather and spatiotemporal features, reducing stock-outs and dead stock while improving turnover.
- Linked customer identities to purchase history over tens of millions of events to power personalized recommendations.
- Shipped a product recognition model for **500+** SKUs used in fast checkout, cutting checkout and queue time.

RESEARCH & INTERNSHIPS

Molecular Dynamics Simulation of Proteins with QM/MM model — Xiamen university, China
(China National Funds) — *Sep. 2014 - Mar. 2017*

- Ran QM/MM protein dynamics simulations and conformational analysis; results published in Acta Cryst. D, PLoS ONE, and Phys. Rev. E (see Publications).

Liquid Biopsy ct-DNA Kit and DNA Methylation Analysis — Lepu Medical, China
(R&D Engineer Intern) — *Mar. 2016 - Mar. 2017*

- Analyzed roughly 10,000 public gene-chip samples (TCGA/GEO) to localize Septin9 methylation sites, an early colon-cancer marker, informing biomarker selection for a clinical diagnostic kit.

AWARDS & SERVICE

- **National First Prize**, China Undergraduate Mathematical Contest in Modeling (CUMCM), 2012.
- **Meritorious Winner**, Mathematical Contest in Modeling (MCM/COMAP), 2013.
- **6th** worldwide, International Solar Decathlon Competition, 2013.
- Ranks **57** and **32** of 1739 teams, IEEE ICDM Global Meteorological AI Challenge, 2018.
- **Second Prize**, 8th CCF Open Source Innovation Competition — Open Source GPU Innovation Ecosystem Regular Season S1, GPU Operator Optimization Challenge (CCF/ODTC), 2025. CCF announcement
- Judge, first screening, Best Practices of Applied Algorithms, 2021.

PUBLICATIONS

- Chen X. GRC-PIM: A Drift-Gated, GEMV-Reusing Residual Calibrator for Soft-Decision Error Correction in Processing-in-Memory. *npj Unconventional Computing* (submitted).
- Cai Y, Su M, Ahmad A, Hu X, Sang J, Kong L, Chen X, Wang C, Shuai J, Han A. Conformational dynamics of the essential sensor histidine kinase WalK. *Acta Crystallographica Section D: Structural Biology.* 2017 Oct 1;73(10):793-803.
- Xia H, Chen X, Chen X, Cheng J, Liu Y, Chen X, Zhang Q, Zou G. Chiral discrimination of amino acid enantiomers based on different interactions with Cu2+. *Sensors and Actuators B: Chemical.* 2018 Jan 1;254:44-51.
- Ahmad A, Cai Y, Chen X, Shuai J, Han A. Conformational dynamics of response regulator RegX3 from Mycobacterium tuberculosis. *PloS one.* 2015 Jul 22;10(7):e0133389.
- Chen Y, Qi H, Li X, Cai M, Chen X, Liu W, Shuai J. Suppressing effect of C a 2+ blips on puff amplitudes by inhibiting channels to prevent recovery. *Physical Review E.* 2016 Aug 15;94(2):022411.

Open Source Architecture Appendix

Chen Xingqiang · github.com/chenxingqiang · zone-based end-to-end system architecture

主数据流　　虚线框 = 架构分区

回退 / 反馈　　盒内 = 模块 + 职责

CONTENTS · one architecture per page

TraceSmith
GPU profiling & replay

YiRage
RuntimeFusion compiler

JAX-sklearn
transparent JAX acceleration

Secret-Learn
privacy-preserving ML

YouTube-DNN
two-tower recall

Research CLI
agent workflow

Questions-Gen
three-stage training

阅读约定

自上而下读取主数据流；虚线框表示架构分区；盒内仅列模块与职责。

蓝色实线表示主链路，紫色虚线表示回退或反馈，不以源码文件充当架构层。

TraceSmith — 系统架构 · GPU Trace 端到端数据流

Application → bindings → capture / state / format → replay & export → GPU backends

主数据流　虚线框 = 架构分区
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APPLICATION · 用户入口

C++ / Python Application
嵌入业务进程

CLI
profile · record · view · export

External UI
Perfetto · Chrome

BINDINGS · Python / C++ 边界

_tracesmith.so
pybind11 native extension

CORE LIBRARY · C++17

Capture Module
CUPTIProfiler · MCPTIProfiler
Metal · Ascend · ROCm · eBPF

Common + Format
RingBuffer SPSC · StackCapture
SBTWriter / SBTReader

State Module
GPUStateMachine
TimelineBuilder · PerfettoExporter

Replay Module
ReplayEngine · StreamScheduler
DeterminismChecker · FrameCapture

CROSS-CUTTING · 集群与调试

replay / debug feedback

Cluster Module
GPUTopology · TimeSync
NCCLTracker

GDB / Tracy
RSPHandler · GPUDebugEngine
TracyExporter

GPU BACKENDS · 系统接口

NVIDIA
CUDA · CUPTI · nsys

MetaX
MACA · MCPTI · mcTracer

Apple / AMD
Metal · xctrace · ROCm

Huawei / Linux
CANN · msprof · eBPF

OUTPUTS · Trace 产物

trace.sbt
compact binary

trace.json / .perfetto
Perfetto timeline

graph.dot / ASCII
analysis views

主链：GPU event → lock-free ring → SBT/state → replay 或 Perfetto；Cluster / GDB 为旁路能力。

YiRage (YPK) — 系统架构 · RuntimeFusion 数据流

Inference engine owns scheduling/KV; RF owns fusion identity; search and execute use the same backend

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INFERENCE ENGINE · 调度与 KV 所有者

vLLM
continuous batch · PagedAttention

SGLang
RadixAttention · extend meta

Decode Metadata
block table · seq lens · SM budget

RUNTIMEFUSION · 每个 decode step

RF.step(inputs, meta)
select 0..N capsules

FusionCapsule
MLP gate_up + down
segment pipeline

Native Fallback
skip → engine native ops

SEARCH & PLAN · offline / warm-up

superoptimize(backend)
KN / TB graph search

FusionPlan / μGraph
profile · fingerprint verify

Persistent Cache
shape bucket restore

Ray Coordinator
optional distributed search

fallback / re-search

BACKEND MANAGER · 编译与执行边界

BackendRegistry
search + execute same backend

KNGraph / TBGraph
transpiler → kernels

yirage.core
Cython runtime

BACKENDS → HARDWARE

CUDA / MACA
NVIDIA GPU · MetaX C500/C550

ROCm / MPS
AMD GPU · Apple Silicon

Ascend / NKI
Huawei 910 · AWS Neuron

CPU / MLIR / Triton
x86 · ARM · extra targets

主链：Engine metadata → RF.step → FusionCapsule → BackendRegistry → target hardware。

JAX-sklearn (xlearn) — 五层架构 · 透明加速数据流

100% sklearn API → runtime injection → proxy dispatch → array conversion → JAX hardware

主数据流　　虚线框 = 架构分区

回退 / 反馈　　盒内 = 模块 + 职责

USER CODE · sklearn-compatible API

import xlearn as sklearn
LinearRegression · KMeans · PCA

Estimator Calls
fit · predict · transform

COMPATIBILITY · xlearn/__init__.py

__getattr__
lazy module loading

_auto_jax_accelerate_module
runtime class injection

AcceleratorRegistry
algorithm implementation map

JAX ACCELERATION · xlearn/_jax

EstimatorProxy
fit-time dispatch

Universal JAX
JIT · vmap · accelerated estimator

sklearn Fallback
original estimator

graceful fallback

DATA MANAGEMENT · boundary conversion

NumPy → JAX
dtype · shape normalization

Policy / Thresholds
dataset size · platform · config

JAX → NumPy
sklearn-compatible outputs

HARDWARE ABSTRACTION · JAX backend

CPU
jaxlib

CUDA GPU
JAX XLA

TPU
JAX XLA

Apple Metal
experimental

代理根据算法、数据规模和配置选择 JAX；不满足条件或执行失败时回退原生 sklearn。

Secret-Learn — 六层架构 · 隐私计算数据流

sklearn-compatible API → privacy mode → JAX / SecretFlow compute → secure devices

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APPLICATION · 多方隐私数据

Healthcare / Finance
cross-party modeling

IoT / Research
distributed learning

SKLEARN API · 统一算法入口

secretlearn.*
fit · predict · transform

algorithm_classifier.py
classify fit signature / epochs

PRIVACY MODES · execution placement

FEDERATED LEARNING
PYU local compute
HEU secure aggregation

SECRET SHARING
SPU MPC · ABY3 / CHEETAH

SPLIT LEARNING
split model · encrypted activations

ACCELERATION & COMPUTE

JAX Acceleration
JIT · vmap · xlearn

SecretFlow Runtime
party orchestration

SECURE DEVICES

PYU
party-local plaintext

HEU
homomorphic aggregation

SPU
secure MPC

TEE
optional enclave

INFRASTRUCTURE & HARDWARE

SecretFlow
distributed secure runtime

JAX / XLA
CPU · GPU · TPU

sklearn
API compatibility

同一 sklearn 调用由 FL / SS / SL 决定数据驻留与执行设备；JAX 加速可公开的本地计算。

YouTube-DNN-RecSys — 系统架构 · 召回数据流

Python / Spark data preparation → TFRecords → dnn.py → user & item vectors → prediction

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DATA PROCESSING · 离线样本构建

Python Pipeline
data2tfrecords.py
read_tfrecords.py

Spark Pipeline
MakeDataOne.scala
MakeDataTwo.scala

FeatureBuilder
history + continuous features

STORAGE · 训练与模型产物

TFRecords
train / evaluation

Checkpoints
ckpt / eval / modelpath

MODEL TRAINING · src/python/models/dnn.py

Input
visit_items_index × 5
continuous_features × 16

Item Embedding
500 × 16 · average pooling

User Tower
Dense 64 → 32 → vector 16

Softmax Output
500 classes · Top-K

evaluation feedback

PREDICTION · serving paths

Python SavedModel
load_dnn_model.py

User Vector
PredictUserVector.scala

Item Table
ItemEmbeddingPredictor.scala

RECOMMENDATION OUTPUT

Candidate Recall
user vector × item embedding

Top-K Items
next-item recommendation

主链：行为历史 → TFRecords → embedding mean-pool → user vector → item table matching → Top-K。

Research CLI — 系统架构 · Agent 请求数据流

Terminal UI → core orchestration & routing → tools / MCP / research modules → model providers

主数据流　回退 / 反馈　虚线框 = 架构分区　盒内 = 模块 + 职责

CLI FRONTEND · packages/cli

Interactive TUI
research.tsx · sessions

Commands
mcp · skills · hooks

ACP
acpSession.ts

CORE ORCHESTRATION · packages/core

Context & Memory
config.ts · memory.ts

modelRouterService
provider selection

Routing Strategy
default · classifier · approval

TOOLS & INTEGRATIONS

Workspace Tools
filesystem · shell

MCP
external service adapters

Research Modules
writing · bibliography · analysis

SDK / A2A / VS Code
embedded clients

tool result loop

MODEL PROVIDERS · provider config

Gemini / Vertex
hosted models

OpenAI / Anthropic
hosted models

Baidu / DeepSeek / Qwen
hosted models

RESULT STREAM

Tool Results
structured observations

Assistant Response
streamed to terminal

主链：prompt → packages/cli → packages/core router → provider / tools → structured result → terminal。

Questions-Gen — 系统架构 · 三阶段训练数据流

Training data → SFT → GRPO → DeepSeek distillation → validation → Hugging Face / Ollama

主数据流　　虚线框 = 架构分区
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TRAINING DATA · problem mixture

Historical Problems
50% source distribution

Variants
30% conditional variation

Novel Problems
20% innovation target

Stage 1 SFT · QuestionsGenTrainer.py

Qwen3-14B
supervised fine-tuning

Stage-1 Checkpoint
stage1-fp-merged

Stage 2 GRPO · reward optimization

RewardCalculator
0.4 difficulty · 0.3 novelty
0.2 rigor · 0.1 diversity

NoveltyConstraint
variation / duplication guard

GRPO Trainer
group-relative optimization

Stage 3 Distill · teacher supervision

quality feedback

DeepSeekTeacher
DeepSeek-R1 reasoning teacher

Final Model
final-merged-16bit

VALIDATION

ModelValidator
single checkpoint

BatchValidator
algebra · geometry · calculus

QualityEvaluator
clarity · difficulty · originality

DEPLOYMENT

Hugging Face
three published stages

Ollama
local inference

questions-gen CLI
validate · batch · quality

主链：混合题库 → SFT → GRPO reward/novelty → DeepSeek-R1 distill → validate → publish。